

■ ENTERPRISE AI GOVERNANCE DECK

AGENTSENTRY

Enterprise Governance Sidecar Proxy & Micro-Telemetry Engine

Real-time interception, token micro-telemetry, and Minto Pyramid executive briefs for financial LLM agent workflows.

🛡️ 100% DETERMINISTIC INTERCEPTION

⚡ REAL-TIME PREFIX CACHING

👤 AUTOMATED PARTNER ESCALATION

Strategic Overview

100%

Deterministic Interception

Captures plurals & syntax shifts with root-stem matching.

25%

Token Cost Savings

Achieved via prefix caching & micro-telemetry tracking.

ZERO

Policy Drift

Structured Minto briefs & real-time partner queues.

⚠ Strategic Context

Autonomous AI agents reduce financial review latency by 80%, but unmonitored LLMs introduce hallucination risks, off-balance sheet overrides, and unconstrained API token spend across client engagements.

🛡 Solution Strategy

AgentSentry acts as an inline sidecar proxy intercepting input scope directives and agent outputs, logging token usage to an immutable SQLite ledger, and escalating high-risk traces to partner queues.

Compliance Vulnerabilities



Syntax Evasion

Keyword matching fails when high-risk terms shift tense or pluralize (e.g., '**liability**' vs '**liabilities**'), allowing critical policy breaches to bypass standard keyword filters.



Opaque Token Spend

Autonomous agent loops call LLM APIs repeatedly without cost boundaries, leading to exponential API spend without node-level latency or token visibility across multi-step agent chains.



Unstructured Outputs

LLMs generate verbose, non-standardized narrative text lacking audit structure, preventing senior decision-makers from rapidly parsing key audit findings or verifying execution states.

Core Framework Components

LangGraph Engine

DAG ORCHESTRATOR

Manages state transitions between financial analyst nodes and brief synthesis compilers in a deterministic directed acyclic graph (DAG).

Compliance Firewall

SIDECAR PROXY

Inspects input directives and node outputs against truncated root-stem arrays (e.g., '**undisclosed** **liabilit**') in real-time.

Telemetry Engine

THREAD-SAFE LEDGER

Logs prompt/completion tokens, node latency, and cost optimizations into an immutable, thread-safe SQLite audit ledger.

Minto Brief Compiler

SYNTHESIS ENGINE

Synthesizes structured briefs (Situation, Complication, Directive) with automated escalation flags for partner approval queues.

Compliance Firewall Logic

▼ Root-Stem Rule Array

Matches outputs against non-evadable truncated root stems:

```
HIGH_RISK_STEMS = [  
  "guarantee", "override", "unverified",  
  "off-balance", "material breach",  
  "revenue bypass", "undisclosed liabilit"  
]
```

📊 Deterministic Scoring Engine

Base Compliance Score	0.10
Weight per Matched Stem	+0.35
Score Ceiling	1.00 MAX
Escalation Threshold	≥ 0.75 (Partner Queue)

State Graph & Monitoring



Node Orchestration

LangGraph **StateGraph** coordinates node transitions. AuditorAnalystNode logs execution metrics, evaluates financial content, and passes state to MintoCompilerNode before reaching END.

Telemetry Interception

At each graph step, latency, prompt tokens, and completion tokens are intercepted before state mutation, providing granular node-by-node cost and safety visibility.

High-Risk Policy Escalation Run

 Execution Context

ESC-REQUIRED

Target: Off-Balance Sheet Entities & Revenue Recognition
Trace ID: tr-88f2-990a
Total Spend: \$0.01430 (Base: \$0.01980)
Savings via Caching: 27.7%

Compliance Risk Index

0.85

 Minto Brief Output

STATUS: ACTION REQUIRED (Escalated to Partner Queue)

Situation: Financial analysis node evaluated off-balance sheet arrangements for Q3 audit.

Complication: Identified 'undisclosed liabilit' and unverified revenue overrides exceeding safety threshold (0.85 >= 0.75).

Directive: Automated pipeline halted. Route trace log to Senior Audit Partner for manual review.

Verified Compliant Run & Audit Trail

✔ Compliant Profile

PASSED

Target: PCAOB ASC 606 Revenue Recognition Review
Trace ID: tr-12a4-44bc
Action: Approved for automated client delivery.

Compliance Risk Index

0.20

🗄 Persistent Ledger Snapshot

```
{
  "AuditorAnalystNode": {"prompt_tokens": 120, "completion_tokens": 250, "cost":
"$0.00652"},
  "MintoCompilerNode": {"prompt_tokens": 180, "completion_tokens": 310, "cost":
"$0.00777"},
  "db_status": "PERSISTED_SQLITE_COMMIT_OK"
}
```


Enterprise ROI & Benefits

Regulatory Risk Defense

100% deterministic interception of policy breaches, eliminating LLM hallucinations and off-balance sheet overrides before output exposure to client teams.

Cost Governance

25%+ immediate reduction in API expenditure via simulated prefix caching and real-time cost limit enforcement across high-volume traces.

Operational Velocity

Sub-second automated escalation of flagged audit tasks directly into senior partner queues with zero policy drift or delay.

Complete Auditability

Immutable, thread-safe SQLite event ledger recording node latency, token usage, and compliance scores for external regulatory compliance.

Deployment Phases

PHASE 1 // COMPLETED

Core Engine & Ledger

- Stemmed Keyword Engine
- Minto Brief Compiler
- Thread-Safe SQLite Ledger
- Core Test Suite

PHASE 2 // NEXT QUARTER

Vector & Multi-Tenant

- Vector Guardrails (Semantic)
- Multi-Tenant SSO & Role Maps
- Redis Cache Intercept Layer
- Partner Portal UI

PHASE 3 // ENTERPRISE SCALE

Infrastructure & SIEM

- Kubernetes Sidecar Mesh
- Splunk / SIEM Integration
- Automated Audit Exports
- Global Multi-Region Sync